MINOR PROJECT REPORT

STUDENT PERFORMANCE ANALYSIS AND PREDICTION SYSTEM

Using Python and Machine Learning

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DECLARATION

I hereby declare that the project entitled **Student Performance Analysis and Prediction System** has been prepared by me as a minor project using Python and machine-learning techniques. The implementation and results described here are based on the project notebook. The dataset is a sample/synthetic dataset and the results are not findings about real students.

ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, **Mr. Pratham Saxena**, and **Shri Shankaracharya Professional University, Bhilai**, for their guidance and support. I also acknowledge the open-source Python libraries and learning resources used in this project.

ABSTRACT

This project demonstrates a Student Performance Analysis and Prediction System using Python. It covers data loading, cleaning, exploratory data analysis, visualization, Linear Regression model training, and evaluation. Study hours per day, attendance percentage, and previous examination marks are used to estimate final examination marks. Performance is measured using MAE, RMSE, R^2, and five-fold cross-validation.

1. INTRODUCTION

Student performance may be associated with study habits, attendance, and previous academic performance. Data analysis and machine learning provide methods to explore such relationships and build predictive models. This project demonstrates a basic end-to-end workflow on a sample dataset.

2. PROBLEM STATEMENT

The objective is to analyze a structured student dataset and build a regression model that estimates final examination marks from selected academic indicators. The system is intended for educational demonstration, not for decisions about real students.

3. OBJECTIVES

- Load and inspect a CSV dataset using Pandas.
- Remove duplicate records and handle missing numerical values.
- Explore the data using descriptive statistics and visualizations.
- Train a Linear Regression model using selected input features.
- Evaluate the model using MAE, RMSE, R^2, and cross-validation.
- Document the workflow, findings, limitations, and future scope.

4. TOOLS AND TECHNOLOGIES

Language: Python
Environment: Google Colab
Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
Algorithm: Linear Regression

5. DATASET DESCRIPTION

The dataset initially contained 301 records and five columns: student_id, study_hours_per_day, attendance_percent, previous_exam_marks, and final_exam_marks. After duplicate removal, 300 rows remained. The three academic indicators were used as input features and final_exam_marks was the target.

6. METHODOLOGY

6.1 Data loading and inspection

The CSV file was loaded into a Pandas DataFrame. The first rows, shape, column information, descriptive statistics, and missing-value counts were inspected.

6.2 Data cleaning

Duplicate records were removed. Missing numerical values, if present, were filled using the mean of the respective column. The cleaned dataset contained 300 rows and five columns.

6.3 Exploratory data analysis

A histogram with a density curve was created to examine the distribution of final examination marks. A correlation heatmap was generated for numerical columns.

6.4 Model development

The input features were study_hours_per_day, attendance_percent, and previous_exam_marks. The target was final_exam_marks. The data was split into training and testing subsets using an 80:20 split and random_state=42. A Linear Regression model was fitted on the training data.

6.5 Evaluation

Predictions were generated for the test set. MAE, RMSE, and R^2 were calculated. Five-fold cross-validation was also performed using R^2 scoring.

7. RESULTS

Metric	Result
Mean Absolute Error (MAE)	6.8797
Root Mean Squared Error (RMSE)	8.2824
Test R^2 score	0.5000
Five-fold R^2 scores	0.5603, 0.4895, 0.4968, 0.5302, 0.6408
Average cross-validation R^2	0.5435

The average cross-validation R^2 was 0.5435, indicating a moderate fit on this sample dataset. R^2 is not the same as percentage prediction accuracy. These results should not be generalized to real educational settings.

8. CONCLUSION

The project implemented a basic machine-learning workflow covering data loading, cleaning, exploratory analysis, visualization, Linear Regression, evaluation, and cross-validation. The model achieved a test R^2 of approximately 0.5000 and an average five-fold cross-validation R^2 of 0.5435 on the sample dataset.

9. LIMITATIONS AND FUTURE SCOPE

- The dataset is synthetic/sample data and may not represent real student populations.
- Only three input features were used.
- Linear Regression may not capture complex non-linear relationships.
- Future work may use a larger, validated dataset with appropriate privacy safeguards.
- Other regression algorithms and feature engineering may be explored.

10. REFERENCES

• Pandas documentation: https://pandas.pydata.org/docs/

• Scikit-learn documentation: https://scikit-learn.org/stable/

• Matplotlib documentation: https://matplotlib.org/stable/

• Seaborn documentation: https://seaborn.pydata.org/

11. APPENDIX – PROJECT SCREENSHOTS

Insert genuine screenshots from the Google Colab notebook in this section before final submission. Recommended screenshots: (1) dataset preview, (2) data-cleaning output, (3) final-marks histogram, (4) correlation heatmap, (5) model training confirmation, (6) MAE/RMSE/R^2 evaluation output, (7) actual-versus-predicted graph, and (8) cross-validation scores. Do not use fabricated screenshots.

Student signature: _____ **Date:** _____

Project Guide signature: _____